PROJECT 4

Deep Visual-Inertial Odometry

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Three networks, shared evaluation

Image pair
(B, 6, 480, 640) · two stacked RGB frames

IMU window
(B, 10, 6) · 3-axis accelerometer + gyroscope

V — Vision

Image pair only

```
VGG8 CNN backbone
512-d feature vector
Heads:  R (6-D continuous)
        t (3-D, metric)
```

SPATIAL SIGNAL

I — Inertial

IMU window only

```
Bidirectional 2-layer LSTM
256-d feature vector
Heads:  R (6-D continuous)
        t (3-D, metric)
```

SCALE SIGNAL

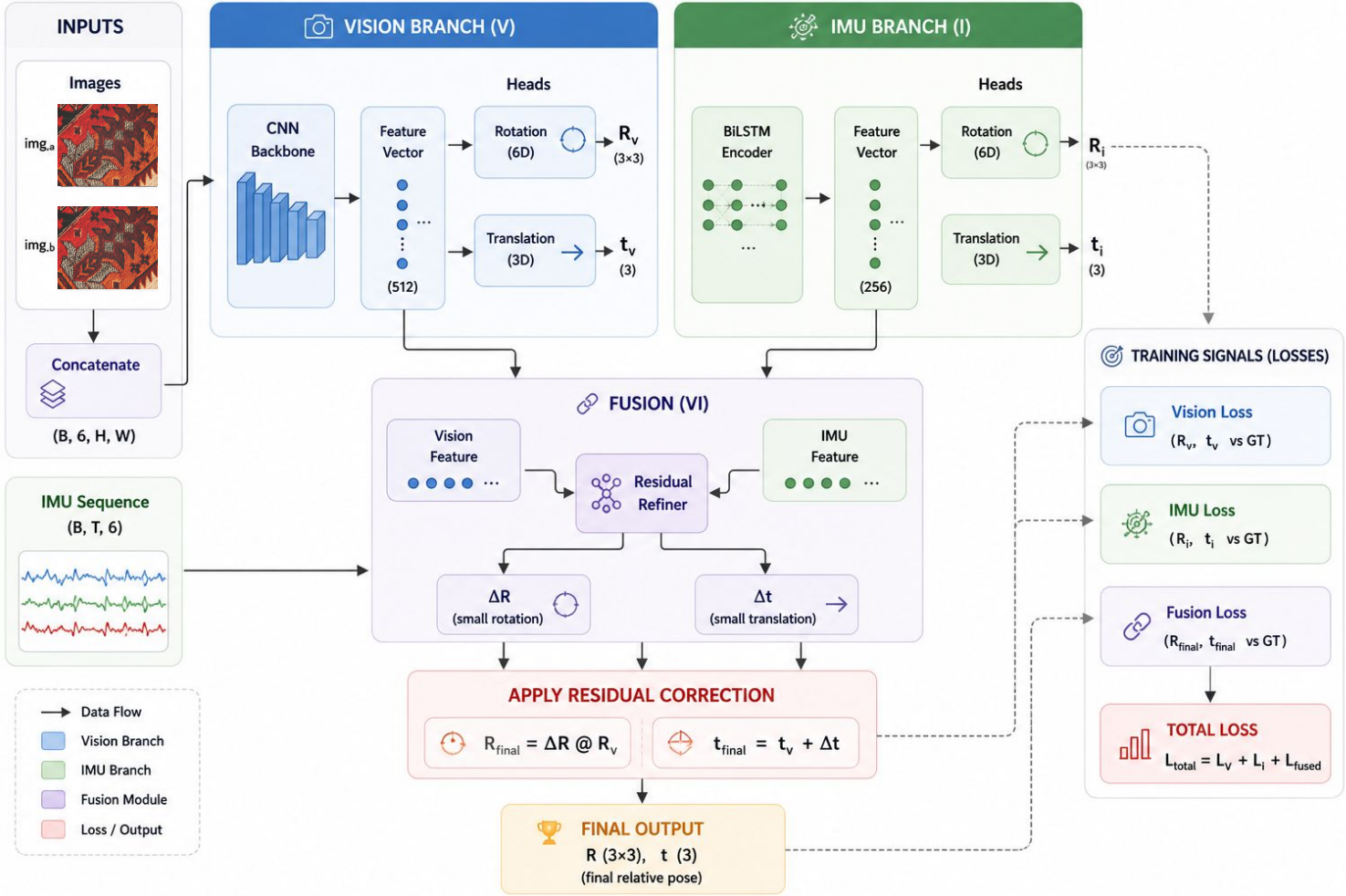
VI — Fusion

Image pair + IMU

```
Residual refiner over
(v_feat, i_feat) concat
R_final = δR · R_v
t_final  = t_v + δt
```

BEST OF BOTH

V predicts pose from images. I predicts from IMU. VI fuses both with a residual refiner.


INPUTS
Images
img_a
img_b
Concatenate
(B, 6, H, W)
IMU Sequence
(B, T, 6)
VISION BRANCH (V)
CNN Backbone
Feature Vector
(512)
Heads
Rotation (6D)
R_v
(3×3)
Translation (3D)
t_v
(3)
IMU BRANCH (I)
BiLSTM Encoder
Feature Vector
(256)
Heads
Rotation (6D)
R_i
(3×3)
Translation (3D)
t_i
(3)
FUSION (VI)
Vision Feature
IMU Feature
Residual Refiner
ΔR
(small rotation)
Δt
(small translation)
APPLY RESIDUAL CORRECTION
R_final = ΔR @ R_v
t_final = t_v + Δt
FINAL OUTPUT
R (3×3), t (3)
(final relative pose)
TRAINING SIGNALS (LOSSES)
Vision Loss
(R_v, t_v vs GT)
IMU Loss
(R_i, t_i vs GT)
Fusion Loss
(R_final, t_final vs GT)
TOTAL LOSS
L_total = L_V + L_i + L_fused
Data Flow
Vision Branch
IMU Branch
Fusion Module
Loss / Output

Loss Function

Three losses, all built from two primitives

Primitives

L_rot — geodesic on SO(3)

$$L_rot(R_pred, R_gt) = \arccos\left(\left(tr(R^{T}_pred \cdot R_gt) - 1\right) / 2 \right)$$

Angle of residual rotation $R^{T}_pred \cdot R_gt$ — shortest rotation from pred to gt.

L_t — L1 / MAE

$$L_t(t_pred, t_gt) = \|t_pred - t_gt\|_1 = \Sigma\square \; |t^{(j)}_pred - t^{(j)}_gt|$$

Sum of per-axis absolute errors; robust to outliers vs L2 / MSE.

w_hover — sigmoid sample reweighting

$$w_hover(\|t_gt\|) = \sigma\left((\|t_gt\| - c) / \nu \right), \quad c = 0.01 \text{ m}, \; \nu = 0.005 \text{ m}$$

Down-weights near-static frames so the network can't "predict zero" to win.

Losses

V branch

L_V — TDirectPoseLoss

geodesic-rotation + sample-reweighted L1-translation

$$L_V = w_R \cdot L_rot(R_v, R_gt) + w_T \cdot \langle \, w_hover \cdot L_t(t_v, t_gt) \, \rangle$$

w_R = 10 *rotation*
w_T = 5 *translation (hover-weighted)*

I branch

L_I — IMUPoseLoss

geodesic-rotation + plain L1-translation (no reweighting)

$$L_I = w_R^{I} \cdot L_rot(R_i, R_gt) + w_T^{I} \cdot \mathbb{E}_b \left[L_t(t_i, t_gt) \right]$$

w_R^{I} = 5 *rotation*
w_T^{I} = 2 *translation (no hover → zero-bias risk)*

VI branch (fused)

L_VI — VILoss_tdirect

auxiliary supervision on each branch + fused-pose supervision

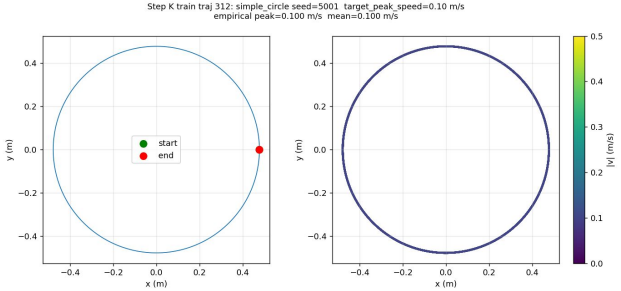
$$L_VI = L_V + L_I + w_R \cdot L_rot(R, R_gt) + w_trans \cdot \mathbb{E}_b \left[L_t(t, t_gt) \right]$$

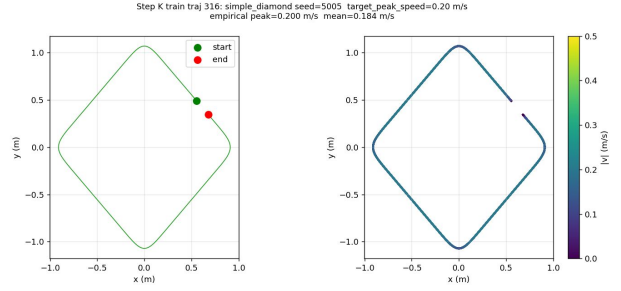
where $(R, t) = (\delta R \cdot R_v, \; t_v + \delta t)$

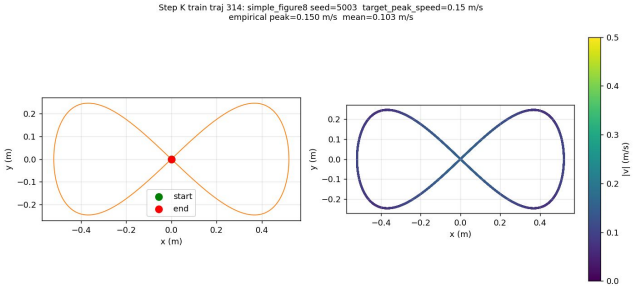
w_R = 10 *fused rotation*
w_trans = 5 *fused translation (no hover)*

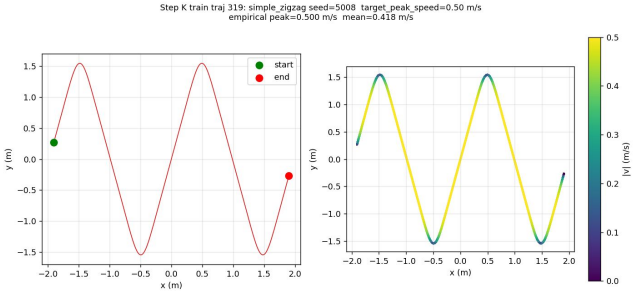
Dataset Generation



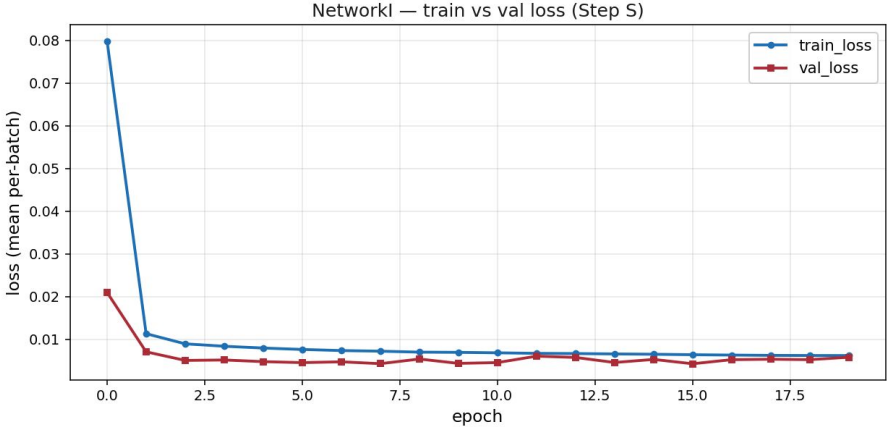


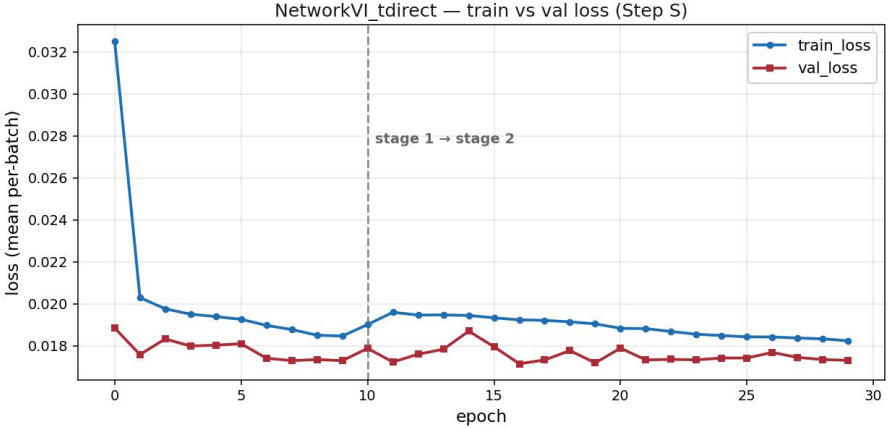


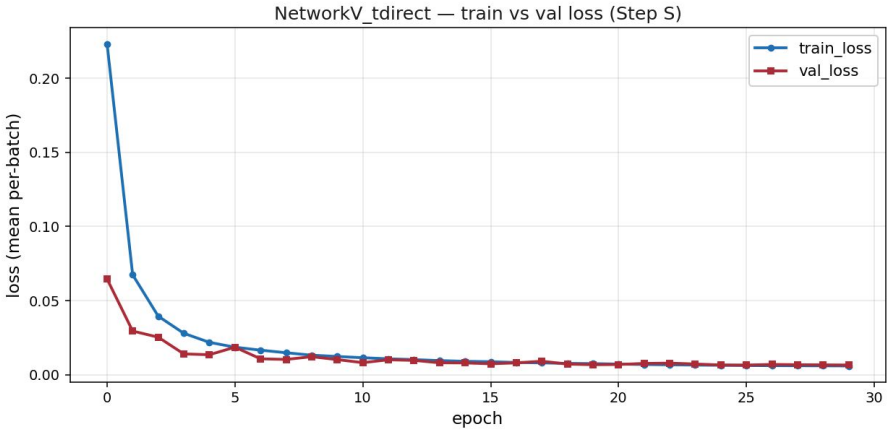




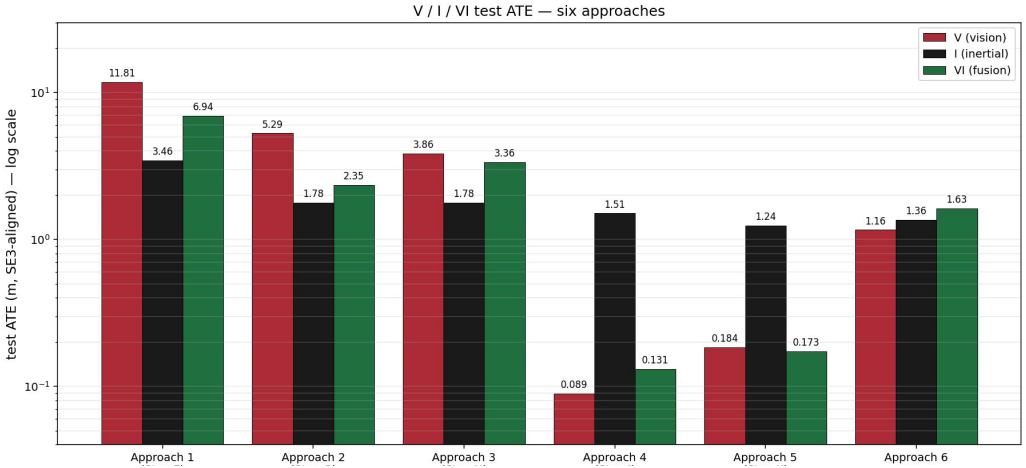
Train Loss

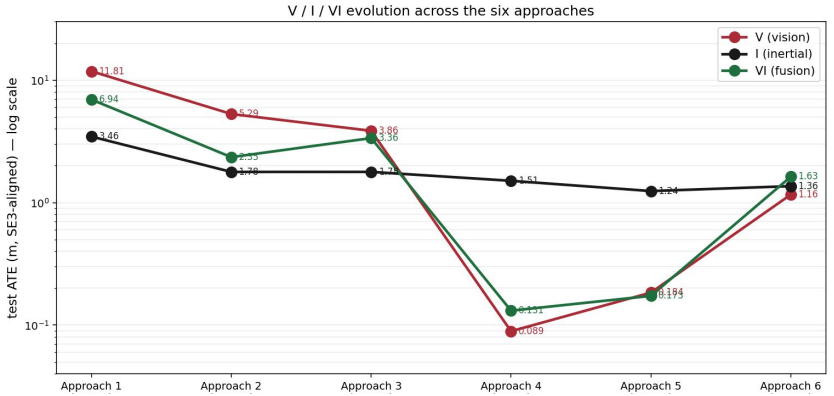

NetworkI — train vs val loss (Step S)
train_loss
val_loss
loss (mean per-batch)
epoch


NetworkVI_tdirect — train vs val loss (Step S)
train_loss
val_loss
stage 1 → stage 2
loss (mean per-batch)
epoch


NetworkV_tdirect — train vs val loss (Step S)
train_loss
val_loss
loss (mean per-batch)
epoch

Results Overview

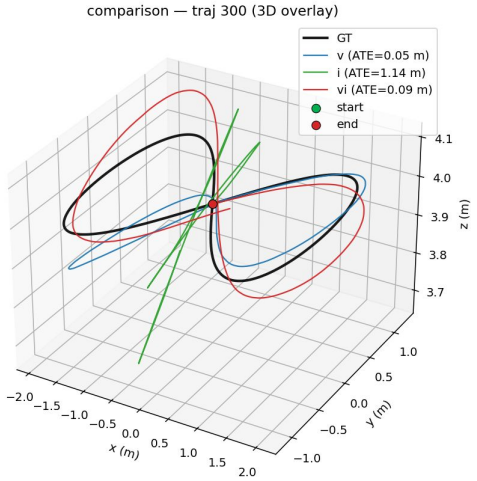


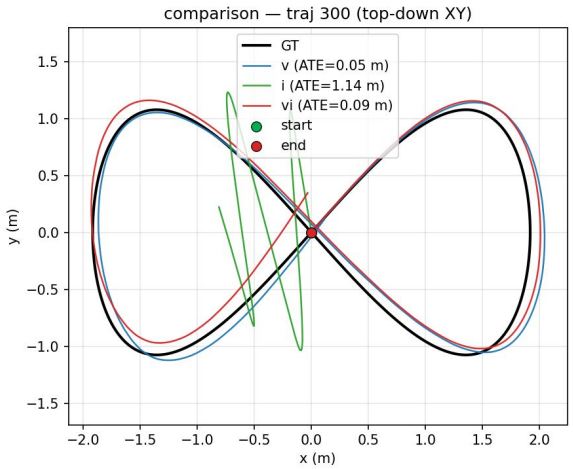


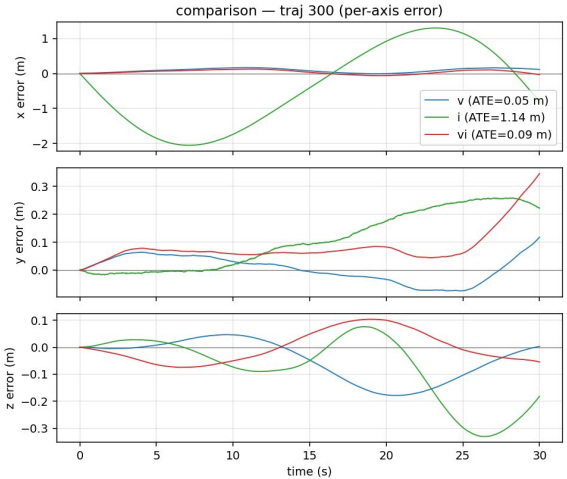
Cumulative ATE across every step (m)

Step	Setup	V	I	VI	What changed
1	Different texture per traj +100 s sinusoid + decomposed head	**11.81**	3.46	6.94	baseline
2	**Fixed texture** + same trajs + decomposed head	**5.29**	1.78	2.35	Texture confound removed → V −55%
3	Fixed texture + **t-direct prediction** (no t/d)	**3.86**	1.78	3.36	Parameterization fixed → V −27%
4	**Simple shapes 30 s + single speed** + 4 train trajs	**0.089**	1.51	0.131	Training-data complexity → V drops 43×
5	Simple shapes + **varied (but slower) speed** + 4 families × 2 speeds	0.184	1.24	**0.173**	Speed diversity → **VI < V** for the first time
6	Step 5 + **corrected IMU pipeline + higher Speed**	1.162	1.362	1.626	I-fix → I monotonically descending from 1.55

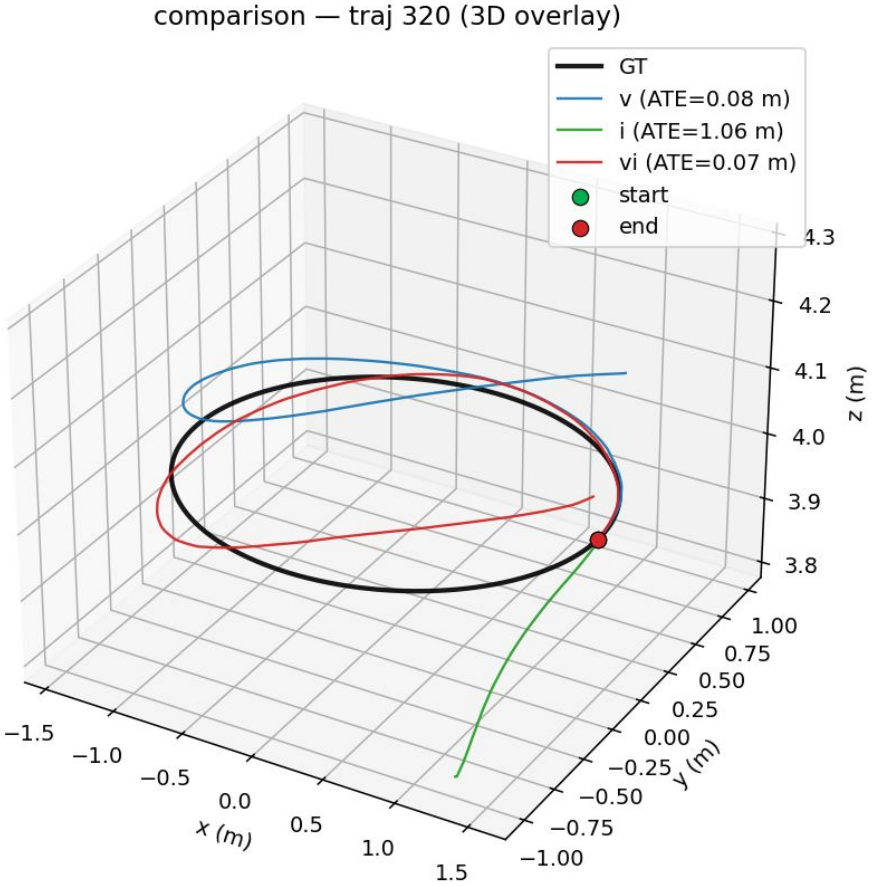
Train Trajectory (GT vs Pred.)

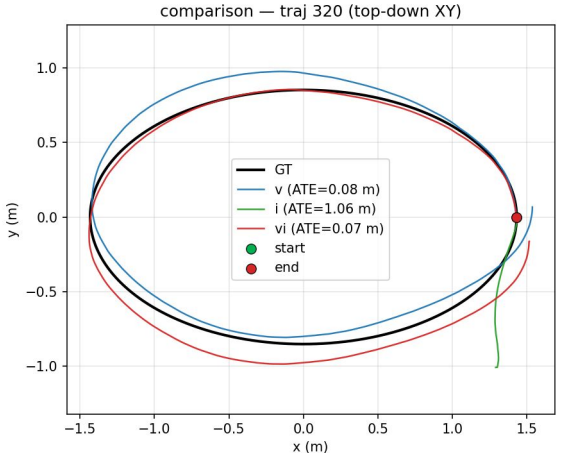


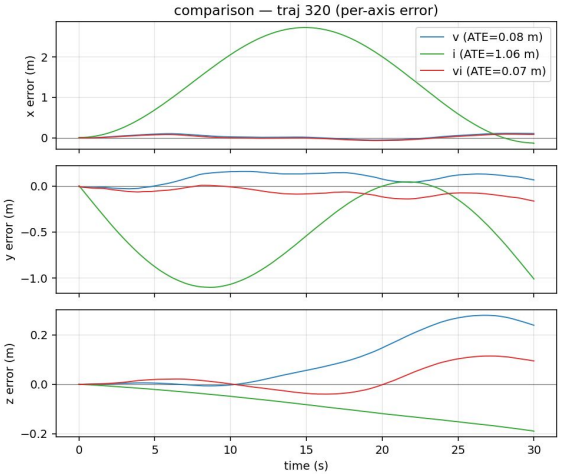




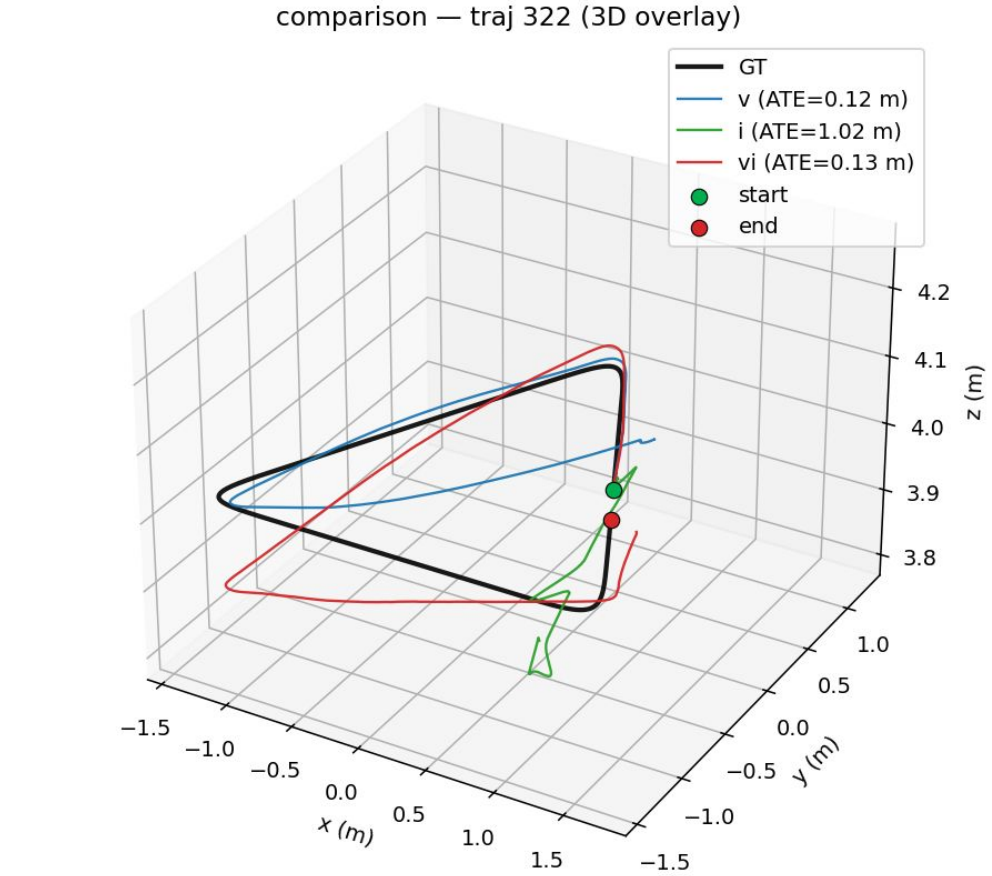
Approach 5

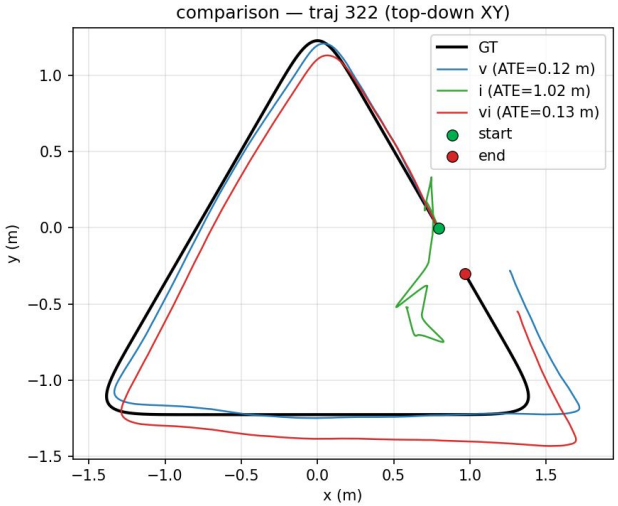

comparison — traj 320 (3D overlay)
GT
v (ATE=0.08 m)
i (ATE=1.06 m)
vi (ATE=0.07 m)
start
end
x (m)
y (m)
z (m)

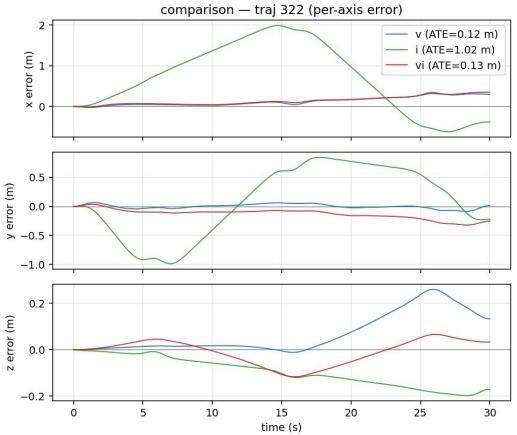

comparison — traj 320 (top-down XY)
GT
v (ATE=0.08 m)
i (ATE=1.06 m)
vi (ATE=0.07 m)
start
end
x (m)
y (m)


comparison — traj 320 (per-axis error)
v (ATE=0.08 m)
i (ATE=1.06 m)
vi (ATE=0.07 m)
x error (m)
y error (m)
z error (m)
time (s)

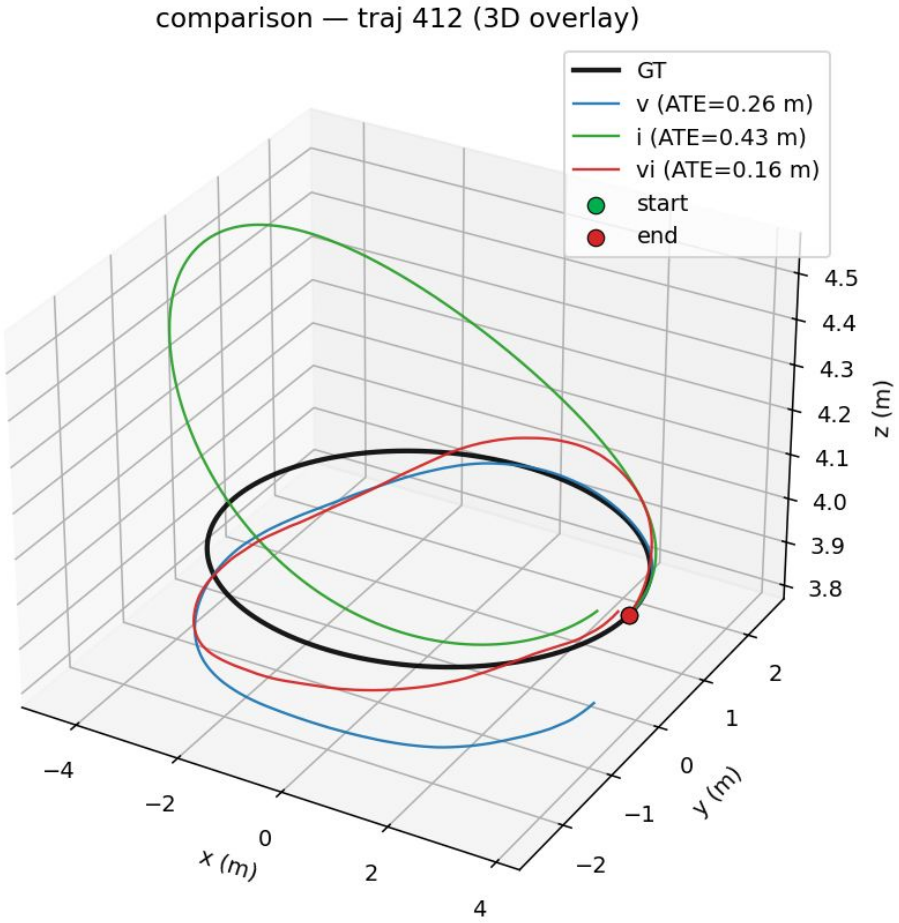
Approach 5

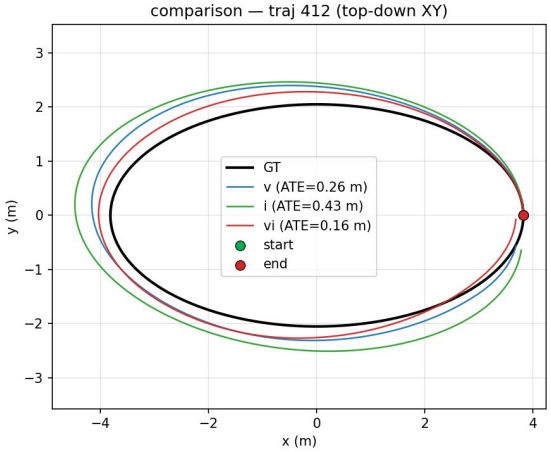

comparison — traj 322 (3D overlay)
GT
v (ATE=0.12 m)
i (ATE=1.02 m)
vi (ATE=0.13 m)
start
end
x (m)
y (m)
z (m)

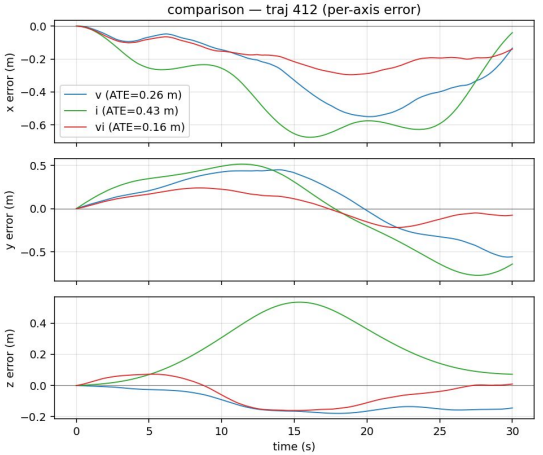

comparison — traj 322 (top-down XY)
GT
v (ATE=0.12 m)
i (ATE=1.02 m)
vi (ATE=0.13 m)
start
end
x (m)
y (m)


comparison — traj 322 (per-axis error)
v (ATE=0.12 m)
i (ATE=1.02 m)
vi (ATE=0.13 m)
x error (m)
y error (m)
z error (m)
time (s)

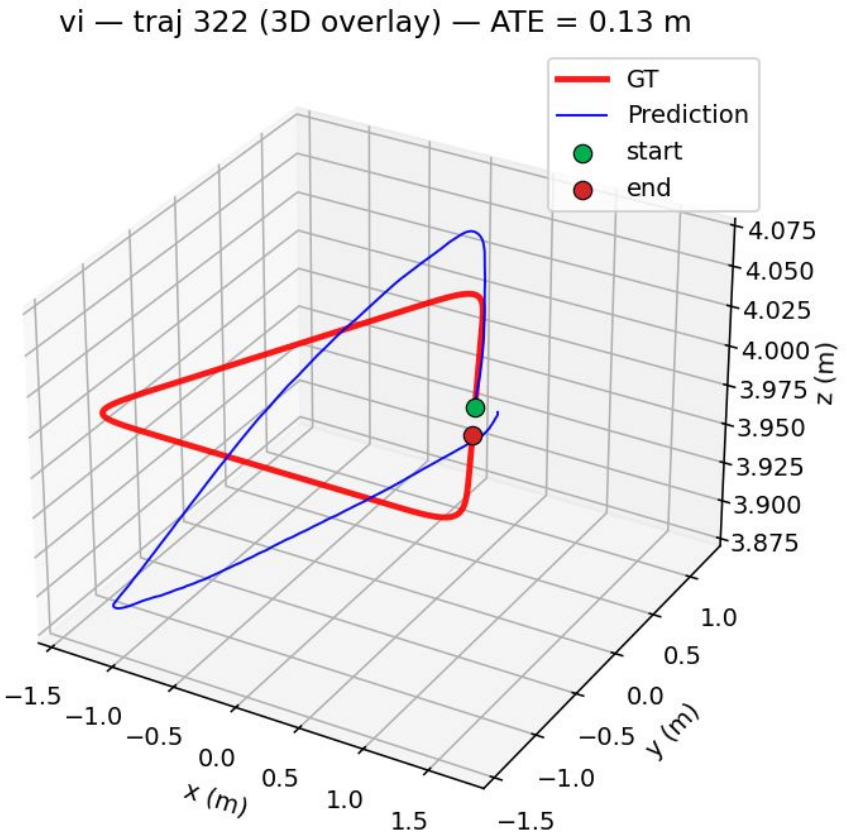
Approach 6 : Ellipse

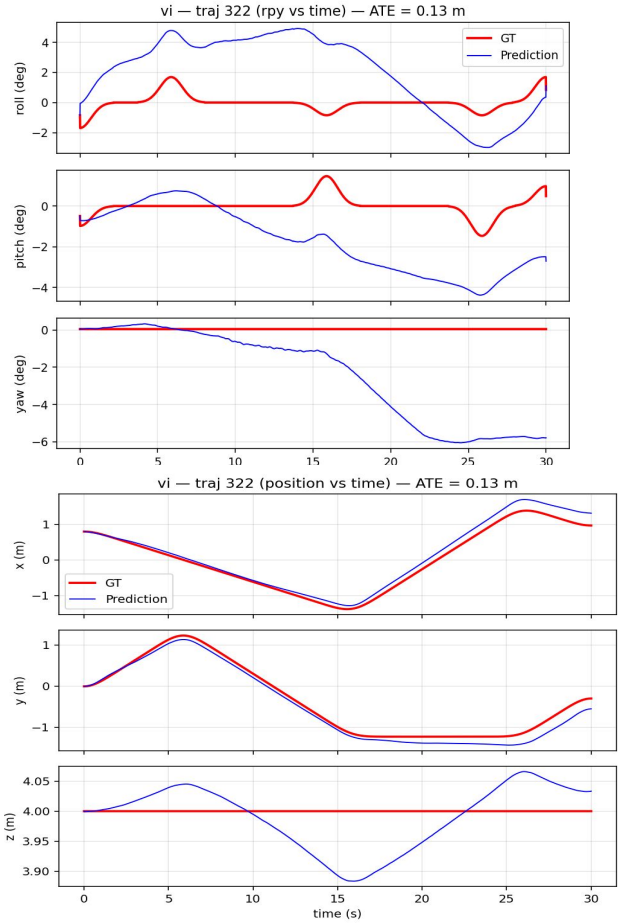






Approach 6 : Triangle





Three confirmed findings

Within-family generalisation works

Speed-diverse training: 0.184 m on novel shapes, 0.078 m on held-out trained-family instances.

Shape novelty alone is fine

0.144 m ATE on a sinusoid_30s shape never seen during training, at controlled speed.

Fusion adds value

VI < V on 4 of 5 test trajs and 6 of 8 train trajs — first time across the project.

Speed Diversity

VI < V for the first time: 0.173 m vs 0.184 m. Variation in motion magnitude was the missing ingredient.

Top 5 Core Limitations

#	Limitation	What it directly causes	Evidence
1	**CNN receptive-field cap on inter-frame motion** VGG8 has no explicit displacement search; feature correspondence breaks when pixel motion exceeds kernel reach (~3–5 px).	V degrades non-recoverably as speed increases.	*V went 0.184 m → 1.16 m moving from Step K (0.3–0.5 m/s) to Step S (0.5–1.5 m/s) with no other change.*
2	**10 ms IMU window — gravity-integration noise floor** Per-window translation signal (~5 mm) is within an order of magnitude of integration noise (~0.5 mm).	I cannot improve below ~1 m mean ATE regardless of encoder upgrades.	*Step S I = 0.43 m at 0.8 m/s (best ever) but I_avg = 1.36 m, worse than Step K's 1.24 m.*
3	**Residual-refiner fusion has no fail-safe** VI = (δR · R_v, t_v + δt); refiner can only nudge V, not override it when V is wrong.	VI degrades V at OOD operating points instead of falling back to I.	*Step S: VI > V on 3 of 4 test trajs (411: V = 0.76 → VI = 2.29).*
4	**Single-frame-pair prediction, no temporal recurrence** Every (R, t) is independent; errors integrate linearly with trajectory length.	Cross-family / long-trajectory generalization is structurally unfixable without architectural change.	*Step K cross-family was a 4–200× scale collapse hidden by SE3 alignment.*
5	**Synthetic-only training** Single texture, single altitude, ideal IMU model, flat ground plane — zero variance on three deployment-relevant axes.	No defined behavior on real video, real IMUs, or multi-depth scenes.	*Step F already showed: changing texture alone broke V (V = 11.81 m on mixed textures).*

What might work

A RAFT / cost-volume vision encoder

+ speed-aware conditioning

Diagnosis

V degraded 0.184 m → 1.16 m as inter-frame displacement grew from 1–2 px (Step K) to 8–10+ px (Step S). VGG8's receptive field can't match features across that gap — there's no explicit displacement search, just convolution.

What to do

- Replace stacked-frame VGG8 with explicit cost-volume encoder (PWC-Net / RAFT-style): per-pixel correlation across ±16 px search window, convex-combine to motion estimate, pool to (R, t).
- Add speed conditioning: feed ‖t_prev‖ as 1-d feature concat'd to global CNN — network learns to scale feature search by expected motion.

Why it's plausible

Step K's RAFT distillation failed because flow magnitudes (0.15–0.80 px) were below RAFT's threshold. At Step S speeds (8–10 px/frame), RAFT is in its training distribution. Re-running the precompute on Step S data alone may flip the negative result with no code changes.

B Curriculum learning + confidence-gated fusion

(my pick if you have one experiment to run)

Diagnosis

Two regressions: (1) V degraded because thrown into 0.5–1.5 m/s from scratch, no slow-speed prior to build on — Step K's working V was discarded. (2) VI worse than V on 3 of 4 trajs — refiner over-corrects using noisy I at OOD points.

What to do

- Curriculum for V: warm-start from Step K checkpoint → +20 ep on 0.1–0.5 m/s → +10 ep adding 0.5–1.5 m/s with lower LR (1e-5). Transfer learning when harder dist is a superset of easier.
- Confidence-gated fusion: V-conf from photometric warp residual, I-conf from gyro-magnitude consistency. α = sigmoid(MLP([conf_v, conf_i, agree(t_v, t_i)])). Soft-blend t and SLERP R.

Why it's plausible

Step K traj 412 (V=0.265, I=0.432, VI=0.161) shows fusion DOES work when both branches have signal — the refiner choking at OOD is the problem, not fusion. Step I's gates collapsed because V was broken (3.86 m); now V has signal at trained speeds, photometric residual will discriminate.